Mathematics for Aerospace Engineers

Chapter 1: Second Order Ordinary Differential Equations

School of Mathematical Sciences

2022–2023

University of Nottingham
UK | CHINA | MALAYSIA

Overview

1 Second order ordinary differential equations
1.1 Introduction

- Linear ODEs arise naturally in the description of physical, chemical and biological systems.
- The aim of this chapter is to familiarise you with basic solution techniques to these equations to enable you to study problems arising in aerospace engineering.
- There is hardly any subject in science or engineering that does not involve differential equations, so these are *extremely important.*

1.2 Definitions

Dependent and independent variables
The full solution to a differential equation expresses the dependent variables in terms of the independent variables. **The dependent variables depend on the independent variables.**

For example, in the ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 1,$$

y is the dependent variable, x is the independent variable, and the solution $y(x) = Ae^{-x} + Bxe^{-x} + 1$ expresses y (the dependent variable) in terms of x (the independent variable).

If a differential equation has more than one independent variable, it is called a *partial differential equation* (PDE). We will study these in depth in Spring.

Order

The order of a differential equation (DE) is that of the highest order derivative appearing in the equation. For example,

$$\frac{\mathrm{d}^4 y}{\mathrm{d}x^4} - \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \sin x$$

is a *fourth order* ODE. In general, an n-th order ODE can be written as

$$F(x, y(x), y'(x), y''(x), \ldots, y^{(n)}(x)) = 0.$$

Linearity

In a *linear* DE, the dependent variable and all its derivatives appear in a linear fashion (i.e. no non-linear functions of the dependent variable or its derivatives appear, and derivatives are not multiplied together with other derivatives or the dependent variable). The most general second order linear ODE is thus

$$a(x)\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + b(x)\frac{\mathrm{d}y}{\mathrm{d}x} + c(x)y = d(x).$$

Any equation for which this is not true, for example

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\frac{q}{L}\sin\theta,$$

is called *nonlinear* ($\sin\theta$ is not linear in θ). Note that $\sin(\theta_1+\theta_2) \neq \sin\theta_1 + \sin\theta_2$.

Homogeneous vs. Inhomogeneous

An ODE is said to be in *standard form* if all terms containing the dependent variable are collected on the left-hand side, and all other terms are on the right-hand side.

A linear ODE is called a *homogeneous* ODE if, in its standard form, the right-hand side is zero. Hence,

$$a(x)\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + b(x)\frac{\mathrm{d}y}{\mathrm{d}x} + c(x)y = 0$$

is the most general second order linear homogeneous ODE.

An ODE that is not homogeneous is called *inhomogeneous*.

In physical terms, an inhomogeneity in a second order linear ODE is a *forcing* term, i.e. an external influence on a physical system.

1.3 Boundary and initial conditions

- In order to find a unique solution to an ODE it is necessary to impose extra conditions on the solution.
- In general, n conditions are required to obtain a unique solution for an nth order ODE.
- One way to do this is to specify the dependent variable and/or some of its derivatives at a particular point.
  - This is called imposing *initial conditions*. For example, a given nth order ODE can be made unique by imposing the initial conditions

    $$y(a) = Y_0, \quad y'(a) = Y_1, \quad \cdots, \quad y^{(n-1)}(a) = Y_{n-1},$$

    specified at some point $x = a$.
  - The ODE, together with the initial conditions, is called an *initial value problem*.
- In aerospace engineering, it is more common to impose conditions at two (or more) points. Such conditions are called *boundary conditions*.

Example 1.3.1
Consider the fourth order ODE

$$\frac{\mathrm{d}^4 y}{\mathrm{d}x^4} - \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \sin x.$$

(a) Imposing the *initial conditions*

$$y(0) = 2, \quad y'(0) = 3/2, \quad y''(0) = 1, \quad y'''(0) = 1/2,$$

leads to the unique solution

$$y(x) = \frac{\sin x}{2} + 1 + e^x.$$

(check by substitution).

(b) Imposing the *boundary conditions*

$$y(0) = 1, \quad y'(0) = 1/2, \quad y(\pi) = 1, \quad y'(\pi) = -1/2,$$

leads to the unique solution

$$y(x) = 1 + \frac{\sin x}{2}.$$

The nature of the unique solution of an ODE depends on the initial/boundary conditions imposed.

1.4 The superposition principle

The superposition principle states that if u_1 and u_2 are two solutions of a *linear homogeneous* ODE, then

$$y = \alpha_1 u_1 + \alpha_2 u_2$$

is also a solution of the same ODE for any constants α_1 and α_2.

For the general homogeneous second order ODE this is easy to show:

$$a(x)\frac{\mathrm{d}^2 u_1}{\mathrm{d}x^2} + b(x)\frac{\mathrm{d}u_1}{\mathrm{d}x} + c(x)u_1 = 0, \qquad (1)$$

$$a(x)\frac{\mathrm{d}^2 u_2}{\mathrm{d}x^2} + b(x)\frac{\mathrm{d}u_2}{\mathrm{d}x} + c(x)u_2 = 0, \qquad (2)$$

Adding (1) multiplied by α_1 to (2) multiplied by α_2 gives

$$a(x)\frac{\mathrm{d}^2}{\mathrm{d}x^2}(\alpha_1 u_1 + \alpha_2 u_2) + b(x)\frac{\mathrm{d}}{\mathrm{d}x}(\alpha_1 u_1 + \alpha_2 u_2) + c(x)(\alpha_1 u_1 + \alpha_2 u_2) = 0,$$

from which it immediately follows that $y = \alpha_1 u_1 + \alpha_2 u_2$ is also a solution of the ODE.

Superposition principle:
If we have two *linearly independent* solutions u_1 and u_2 of a homogeneous linear second order ODE (i.e. solutions that are not just multiples of each other), then *every* solution of that ODE can be written as

$$y = \alpha_1 u_1 + \alpha_2 u_2,$$

for arbitrary constants α_1 and α_2. This is called the *general solution* of the ODE.

Example 1.4.1
Consider the ODE
$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - y = 0.$$
We can easily check that
$$u_1(x) = e^x, \quad \text{and} \quad u_2(x) = e^{-x}$$
are solutions of the ODE. Thus, the general solution is
$$y(x) = \alpha_1 e^x + \alpha_2 e^{-x}.$$
The question is, in general, how do we find two fundamental solutions u_1 and u_2? In this chapter, we will only do this for the most important type of second order ODE.

1.5 Constant coefficient, linear second order ODEs

We consider linear second order ODEs where the coefficients are constant, i.e. do not depend on x. These ODEs have the general form

$$a\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = f(x),$$

where a, b and c are given constants.

- If $f(x) = 0$, the ODE is *homogeneous*. We consider these first.
- Other notation is possible. For example, the ODE might be in terms of $x(t)$ and its derivatives with respect to t.

1.5.1 Homogeneous linear second order ODEs with constant coefficients

To find the general solution of the ODE

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0,$$

we “guess” solutions of the type $y = Ae^{mx}$, where A and m are constants.
Then

$$\frac{dy}{\mathrm{d}x} = Ame^{mx}, \quad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = Am^2e^{mx}.$$

Substituting this into the ODE, we obtain

$$(am^2 + bm + c)Ae^{mx} = 0.$$

As Ae^{mx} cannot be zero, we must have either $A = 0$ (the *trivial solution*), or, for non-trivial solutions,

$$\boxed{am^2 + bm + c = 0.}$$

This quadratic equation in m is called the *auxiliary equation.*
In general, there are two solutions to the auxiliary equation. Using the quadratic formula,

$$m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

So we have two solutions to the ODE,

$$u_1(x) = e^{m_1 x} \quad \text{and} \quad u_2(x) = e^{m_2 x}.$$

Using the superposition principle, the general solution is a combination of these,

$$y(x) = A_1 e^{m_1 x} + A_2 e^{m_2 x},$$

where A_1 and A_2 are constants.

Example 1.5.1
Find the general solution of the ODE

$$\frac{d^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 0.$$

Solution
Let $y = Ae^{mx}$. Here, we have $a = 1$, $b = -1$, $c = -2$, so the auxiliary equation is

$$m^2 - m - 2 = 0 \Rightarrow (m+1)(m-2) = 0.$$

Hence, $m_1 = -1$, $m_2 = 2$.
The general solution for $y(x)$ is

$$y(x) = A_1 e^{-x} + A_2 e^{2x}$$

for arbitrary constants A_1 and A_2.

1.6.1 Three types of solution of the auxiliary equation

Since m_1 and m_2 are roots of a quadratic equation, three different cases are possible:

1. m_1 and m_2 are *real and distinct* (as in the previous example).
2. m_1 and m_2 are *real and repeated* (i.e. $m_1 = m_2$).
3. m_1 and m_2 are *complex conjugates*.

Case 1: real and distinct roots

We have already dealt with Case 1 in the previous example. If m_1 and m_2 are the real distinct roots of the auxiliary equation, the general solution of the ODE is given by

$$\boxed{y(x) = A_1 e^{m_1 x} + A_2 e^{m_2 x}.}$$

Case 2: real and repeated roots
If $m_1 = m_2$, the general solution would appear to be

$$y(x) = A_1 e^{m_1 x} + A_2 e^{m_2 x} = B e^{m_1 x},$$

where $B = A_1 + A_2$. The solution now only contains one constant, when we know it should contain two. In fact, when $m_1 = m_2$, the general solution is

$$\boxed{y(x) = (A_1 + A_2 x) e^{m_1 x}.}$$

Example 1.6.2
Find the general solution of $y'' - 6y' + 9y = 0$.
Solution
Let $y = Ae^{mx}$. Then the auxiliary equation is

$$m^2 - 6m + 9 = 0 \Rightarrow (m-3)^2 = 0 \Rightarrow m_1 = m_2 = 3.$$

Hence, the general solution of the ODE is

$$y(x) = (A_1 + A_2 x) e^{3x}.$$

Case 3: complex conjugate roots
Suppose the roots of the auxiliary equation are

$$m_1 = \alpha + i\beta, \quad m_2 = \alpha - i\beta.$$

The general solution is now

$$\begin{aligned} y &= A_1 e^{\alpha+i\beta} x + A_2 e^{\alpha-i\beta} x, \\ &= A_1 e^{\alpha x} e^{i\beta x} + A_2 e^{\alpha x} e^{-i\beta x}, \\ &= e^{\alpha x}(A_1 e^{i\beta x} + A_2 e^{-i\beta x}), \\ &= e^{\alpha x}[(A_1 + A_2)\cos(\beta x) + i(A_1 - A_2)\sin(\beta x)], \\ &= \boxed{e^{\alpha x}[B_1 \cos(\beta x) + B_2 \sin(\beta x)],} \end{aligned}$$

where B_1 and B_2 are arbitrary constants (they are in fact combinations of A_1 and A_2 but that is irrelevant since A_1 and A_2 are also arbitrary). This is the general solution in this case.

Example 1.6.3
Find the general solution of

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 0.$$

Solution
Let $y = Ae^{mx}$. Then the auxiliary equation is $m^2 + 2m + 5 = 0$, which has complex roots

$$m_1 = -1 + 2i, \quad m_2 = -1 - 2i.$$

Hence, the general solution of the ODE is

$$y(x) = e^{-x}[A_1 \cos(2x) + A_2 \sin(2x)].$$

Summary: solutions of homogeneous second order ODEs

The *general solution* of the ODE

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0$$

is found by solving the *auxiliary equation* $am^2 + bm + c = 0$. Then

1. $\boxed{A_1 e^{m_1 x} + A_2 e^{m_2 x}}$
   if m_1 and m_2 are real distinct roots of the auxiliary equation.
2. $\boxed{(A_1 + A_2 x)e^{m_1 x}}$
   if $m_1 = m_2$ are real repeated roots of the auxiliary equation.
3. $\boxed{e^{\alpha x}[A_1 \cos(\beta x) + A_2 \sin(\beta x)]}$
   if $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$ are complex conjugate roots of the auxiliary equation.

1.7 Inhomogeneous linear second order ODEs with constant coefficients

The linear ODE

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = f(x)$$

is said to be *inhomogeneous*, i.e. there is a term involving x **but not** y on the right-hand side.

This ODE is solved in two stages:

1. Consider the equivalent homogeneous equation obtained by setting $f(x) = 0$,

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0.$$

   Solve this using the previous method.

   This solution is called the *complementary function* (C.F.) written $y_{\mathrm{CF}}(x)$, say.

2. Obtain (for example by trial and error) a *particular solution* to the original inhomogeneous ODE. We call this the *particular integral* (P.I.), denoted $y_{\mathrm{PI}}(x)$, say.

   We need to "guess" a particular integral to see if it works. Usually, the form of the function $f(x)$ gives us some idea of the form of y_{PI} must have.
   We can use our experience of derivatives of functions to help find y_{PI}. For example, differentiating:

   - e^{px} gives pe^{px} (use if $f(x)$ is an exponential).
   - x^n gives nx^{n-1} (use if $f(x)$ is a polynomial).
   - $\sin(px)$ gives $p\cos(px)$ (use if $f(x)$ is $\sin$ or $\cos$).
   - $\cos(px)$ gives $-p\sin(px)$ (use if $f(x)$ is $\sin$ or $\cos$).

   The complete general solution is then

$$\boxed{y(x) = y_{\mathrm{CF}}(x) + y_{\mathrm{PI}}(x)}.$$

Example 1.7.1

Find the general solution of the ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = e^{-x}.$$

1. First solve the homogeneous equation, which has the auxiliary equation $m^2 - 3m + 2 = 0$, which has solutions $m_1 = 1$, $m_2 = 2$. Hence, the complementary function is

$$y_{\mathrm{CF}}(x) = A_1 e^x + A_2 e^{2x}.$$

2. Now seek a particular integral related to $f(x) = e^{-x}$. Try a solution of the form

$$y_{\mathrm{PI}}(x) = Be^{-x}.$$

We have

$$y'_{\mathrm{PI}}(x) = -Be^{-x}, \quad y''_{\mathrm{PI}}(x) = Be^{-x}.$$

Substituting into the ODE,

$$Be^{-x} + 3Be^{-x} + 2Be^{-x} = e^{-x} \Rightarrow 6Be^{-x} = e^{-x}.$$

Equating the coefficients of e^{-x} we obtain $B = 1/6$. Hence, the particular integral is

$$y_{\mathrm{PI}}(x) = \frac{1}{6}e^{-x}.$$

Finally, the general solution of the ODE is

$$\begin{aligned} y(x) &= y_{\mathrm{CF}}(x) + y_{\mathrm{PI}}(x), \\ &= A_1 e^{x} + A_2 e^{2x} + \frac{1}{6}e^{-x}. \end{aligned}$$

Example 1.7.2

Find the general solution of the ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 3 - 2x^2.$$

1. The left-hand side is the same as in the previous example, so the complementary function is the same.
2. Since $f(x) = 3 - 2x^2$ in this case, try a general quadratic particular integral:

$$y_{\mathrm{PI}}(x) = ax^2 + bx + c.$$

We have

$$y'_{\mathrm{PI}}(x) = 2ax + b, \quad y''_{\mathrm{PI}}(x) = 2a.$$

Substituting into the ODE,

$$2a - 3(2ax + b) + 2(ax^2 + bx + c) = 3 - 2x^2,$$
$$\Rightarrow 2ax^2 + (2b - 6a)x + (2a - 3b + 2c) = 3 - 2x^2$$

Equating coefficients:

$$\begin{aligned} x^2: &\quad 2a = -2 \;\Rightarrow\; \boxed{a = -1.} \\ x^1: &\quad 2b - 6a = 0 \;\Rightarrow\; \boxed{b = -3.} \\ x^0: &\quad 2a - 3b + 2c \;\Rightarrow\; \boxed{c = -2.} \end{aligned}$$

Hence, the particular integral is

$$y_{\mathrm{PI}}(x) = -x^2 - 3x - 2.$$

Finally, the general solution of the ODE is

$$y(x) = A_1 e^x + A_2 e^{2x} - x^2 - 3x - 2.$$

Example 1.7.3

Find the general solution of the ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = \sin x.$$

1. Again, the complementary function is the same.
2. Try to find a particular integral of the form $y_{\mathrm{PI}}(x) = a \sin x$, i.e.

$$y'_{\mathrm{PI}}(x) = a \cos x, \quad y''_{\mathrm{PI}}(x) = -a \sin x,$$

Substituting into the ODE ,

$$-a \sin x - 3a \cos x + 2a \sin x = \sin x.$$

Collecting terms, $a \sin x - 3a \cos x = \sin x$.

There is no value of a that satisfies both of

$$a \sin x = \sin x \quad \text{and} \quad -3a \cos x = 0,$$

so we need to try another form of particular integral.

Instead, try a particular integral of the form

$$y_{\mathrm{PI}}(x) = a\sin x + b\cos x.$$

We have

$$y'_{\mathrm{PI}}(x) = a\cos x - b\sin x, \quad y''_{\mathrm{PI}}(x) = -a\sin x - b\cos x.$$

Substituting into the ODE,

$$-a\sin x - b\cos x - 3a\cos x + 3b\sin x + 2a\sin x + 2b\cos x = \sin x.$$
$$\Rightarrow (a+3b)\sin x + (b-3a)\cos x = \sin x.$$

Equating coefficients:

$\cos x$: $\quad b - 3a = 0 \;\Rightarrow\; b = 3a.$

$\sin x$: $\quad a + 3b = 1 \;\Rightarrow\; a + 9a = 1 \;\Rightarrow\; 10a = 1.$

Solving for a and b, we have $a = 1/10$, $b = 3/10$. Hence, the general solution of the ODE is

$$y(x) = A_1e^x + A_2e^{2x} + \frac{1}{10}\sin x + \frac{3}{10}\cos x.$$

Example 1.7.4

This method fails if $f(x)$ has the same form as part (or all) of the complementary function. Consider the ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = e^x.$$

Here, $f(x) = e^x$ appears in the complementary function $y_{\mathrm{CF}}(x) = A_1 e^x + A_2 e^{2x}$. If we choose $y_{\mathrm{PI}}(x) = ae^x$ and substitute into the ODE we obtain

$$ae^x - 3ae^x + 2ae^x = e^x \;\Rightarrow\; 0 = 1???.$$

The solution is to try a particular integral of the form $y_{\rm PI}(x) = axe^x$. Then (using the product rule),

$$y'_{\rm PI}(x) = axe^x + ae^x, \quad y''_{\rm PI}(x) = axe^x + 2ae^x.$$

Substituting into the ODE,

$$\begin{aligned} axe^x + 2ae^x - 3(axe^x + ae^x) + 2axe^x &= e^x, \\ \Rightarrow -ae^x = e^x \;\Rightarrow\; a = -1. \end{aligned}$$

Finally, the general solution of the ODE is

$$y(x) = A_1 e^x + A_2 e^{2x} - xe^x.$$

Example 1.7.5 Find the particular solution of the ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} = e^x,$$

subject to the initial conditions

$$y(0) = 1 \quad \text{and} \quad y'(0) = 0.$$

In the previous example, we found that the general solution of this ODE is

$$y(x) = A_1 e^x + A_2 e^{2x} - xe^x.$$

We use the initial conditions to find A_1 and A_2. At $x = 0$, we have

$$y = A_1 + A_2 = 1 \quad \text{and} \quad y' = A_1 + 2A_2 - 1 = 0.$$

Hence, $A_1 = 1$ and $A_2 = 0$, so the particular solution of the ODE with these initial conditions is

$$y = e^x - xe^x.$$

Summary: choosing the particular integral

- If $f(x)$ is a polynomial of degree n, choose the most general polynomial of degree n, e.g. if $n = 2$, choose $y_{\mathrm{PI}}(x) = ax^2 + bx + c$.
- If $f(x)$ is $e^{\alpha x}$, choose
$$y_{\mathrm{PI}}(x) = Be^{\alpha x}.$$
- If $f(x)$ is $\cos(\alpha x)$ or $\sin(\alpha x)$, choose
$$y_{\mathrm{PI}}(x) = a\cos(\alpha x) + b\sin(\alpha x).$$

These are the only forms of $f(x)$ that will appear in this module. Of course, you may encounter ODEs with more complicated right-hand sides and so will need to “guess” a more sophisticated particular integral.

You will get better at this with practise!

1.8 Systems of first order linear ODEs

Consider a second order ODE, e.g.

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = x.$$

This can be written as two coupled first order ODEs for two dependent variables u and y as follows:

Define $u = \dfrac{\mathrm{d}y}{\mathrm{d}x}$, then $\dfrac{\mathrm{d}u}{\mathrm{d}x} + 3u + 2y = x$.

This system of two first order ODEs is equivalent to the original second order ODE.

Similarly, a system of first order ODEs can be rewritten as a single second order ODE.

Example 1.8.1
Consider the system of first order ODEs

$$\frac{\mathrm{d}y}{\mathrm{d}x} + u = 3, \tag{3}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} + y = 2. \tag{4}$$

To solve this system, first differentiate (3) to give

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{\mathrm{d}u}{\mathrm{d}x} = 0,$$

then use (4) to eliminate $\mathrm{d}u/\mathrm{d}x$:

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - y + 2 = 0.$$

This is called the *method of elimination*. The second order ODE above can be solved using the methods we studied previously. The solution is

$$y(x) = A_1 e^x + A_2 e^{-x} + 2. \quad \text{(Check this!)}$$

The solution for u can now be found directly from (3):

$$u(x) = 3 - \frac{\mathrm{d}y}{\mathrm{d}x} = 3 - A_1 e^x + A_2 e^{-x}.$$

The constants A_1 and A_2 can be found using initial/boundary conditions.

For example, if $y(0) = 1$, $u(0) = 0$, then $A_1 + A_2 + 2 = 1$ and $3 - A + B = 0$, giving $A_1 = 1$, $A_2 = -2$. Hence, the particular solution to the system for these initial conditions is

$$y(x) = e^x - 2e^{-x} + 2,$$
$$u(x) = 3 - e^x - 2e^{-x}.$$

Example 1.8.2

Find the particular solution to the system of ODEs

$$\frac{\mathrm{d}x}{\mathrm{d}t} + 2x + y = 0, \quad \frac{\mathrm{d}y}{\mathrm{d}t} - 3x - 2y = e^{-2t},$$

subject to the initial conditions $x(0) = 0$, $y(0) = 0$.

Differentiating the first equation,

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + 2\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\mathrm{d}y}{\mathrm{d}t} = 0.$$

Substituting for $\frac{\mathrm{d}y}{\mathrm{d}t}$ using the second equation,

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + 2\frac{\mathrm{d}x}{\mathrm{d}t} + 3x + 2y + e^{-2t} = 0.$$

We eliminate y by rearranging the first equation:

$$y = -\frac{\mathrm{d}x}{\mathrm{d}t} - 2x.$$

Hence,

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} - x = -e^{-2t}.$$

The auxiliary equation for this second order ODE is $m^2 - 1 = 0$, which has real distinct roots $m_1 = 1$, $m_2 = -1$. Hence, the complementary function is

$$x_{\mathrm{CF}}(t) = A_1 e^t + A_2 e^{-t}.$$

For the particular integral, try $x_{\mathrm{PI}}(t) = Be^{-2t}$. Substituting into the ODE,

$$4Be^{-2t} - Be{-}2t = -e^{-2t},$$

$$\Rightarrow 3B = -1 \;\Rightarrow\; B = -\frac{1}{3}.$$

The general solution for $x(t)$ is

$$x(t) = A_1 e^t + A_2 e^{-t} - \frac{1}{3} e^{-2t}.$$

Using the first equation,

$$\begin{aligned} y(t) &= -\frac{\mathrm{d}x}{\mathrm{d}t} - 2x, \\ &= -A_1 e^{t} + A_2 e^{-t} - \frac{2}{3}e^{-2t} - 2A_1 e^{t} - 2A_2 e^{-t} + \frac{2}{3}e^{-2t}, \\ &= -3A_1 e^{t} - A_2 e^{-t}. \end{aligned}$$

Applying the initial conditions,

$$\begin{aligned} x(0) &= A_1 + A_2 - \frac{1}{3} = 0, \\ y(0) &= -3A_1 - A_2 = 0. \end{aligned}$$

Solving these yields $A_1 = -1/6$, $A_2 = 1/2$. Hence, the particular solution of the system is

$$x(t) = -\frac{1}{6}e^{t} + \frac{1}{2}e^{-t} - \frac{1}{3}e^{-2t}, \quad y(t) = \frac{1}{2}e^{t} - \frac{1}{2}e^{-t}.$$